

FYP-I Project Proposal

OmniTrack AI: A Secure AI-Driven Suite for Multi-Camera Retail Analytics and Behavioral Integrity

Session: **SPRING 2026**
Degree Program: **BS Cyber Security**
Section: **[SECTION]**
Batch: **[BATCH]**

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This proposal follows the FYP-I Project Proposal template for Session SPRING 2026.

Project Abstract

OmniTrack AI is a professional-grade retail analytics platform engineered to transform standard surveillance feeds into actionable business intelligence while maintaining a hardened security posture. In modern retail environments, CCTV cameras serve primarily as passive security tools with minimal contribution to operational insights. OmniTrack addresses this gap by deploying State-of-the-Art (SOTA) Computer Vision techniques, specifically the Ultralytics framework (YOLOv8/v11) for real-time person detection, PyTorch-based Torchreid for Global Re-Identification (Re-ID), and a suite of specialized analytics modules -- including video synopsis generation, shelf engagement and top-selling product detection, fire and smoke detection for safety compliance, crowd density estimation, checkout counter analytics, and customer emotion recognition.

Deep learning feature vectors (128-d or 512-d embeddings) are extracted from each detected person and stored in a pgvector-enhanced PostgreSQL database, enabling high-speed cosine similarity searches with sub-100ms retrieval latency for cross-camera identity matching.

Beyond customer tracking, the platform provides a comprehensive operational intelligence layer. A Video Synopsis module condenses hours of surveillance footage into compact, reviewable summaries by extracting and compositing only the segments containing moving objects. Shelf-facing cameras analyze product interaction zones to identify top-selling areas. A dedicated fire and smoke detection module ensures real-time safety monitoring with instant alerting. Crowd density estimation classifies zones as crowded or uncrowded. Checkout counter analytics track queue lengths, service times, and order throughput. A facial emotion recognition module leveraging DeepFace/FER gauges customer sentiment across the store.

The project follows a phased development approach across two semesters. FYP-1 focuses on building and validating the core AI modules -- including detection, tracking, Video Synopsis processing, Global Re-ID model training, and all extended analytics modules -- while FYP-2 delivers the fully integrated, production-ready end-to-end system.

To satisfy the demands of a BS Cybersecurity curriculum, the project implements a Decoupled Security Pipeline with SHA-256 hashing for tamper-evident audit trails, AES-256 encryption via PyCryptodome, and adversarial robustness evaluation using the Adversarial Robustness Toolbox (ART). The backend is powered by FastAPI, while the frontend delivers real-time dashboards covering all analytics modules. By integrating high-performance AI with rigorous cryptographic standards, OmniTrack AI provides a resilient, end-to-end solution for retail optimization and behavioral integrity verification.

1. Introduction

The retail industry is undergoing a digital transformation where physical operations increasingly rely on data-driven decision-making. Despite the widespread deployment of CCTV infrastructure in retail stores, the vast majority of video feeds remain underutilized -- serving only as passive recording devices for post-incident review. This represents a significant missed opportunity for extracting real-time operational intelligence such as customer flow patterns, dwell-time analytics, zone-level engagement metrics, safety monitoring, and customer sentiment analysis.

OmniTrack AI aims to bridge the gap between physical retail operations and digital data analytics by developing a multi-camera intelligence suite capable of identifying and tracking individuals throughout a store environment. Beyond person tracking, the system provides a holistic operational view by generating compact video synopses for rapid footage review, analyzing shelf engagement, detecting safety hazards, monitoring crowd density, tracking checkout efficiency, and recognizing customer emotions.

Main Sub-tasks:

- **Detection & Tracking:** Utilizing the Ultralytics framework (YOLOv8/v11) combined with ByteTrack for real-time person detection and multi-object tracking (MOT) across video frames from multiple camera feeds.
- **Global Re-Identification (Re-ID):** Employing the Torchreid library (PyTorch) to extract 128-d or 512-d feature embeddings for cross-camera identity matching without facial recognition. (Processing begins in FYP-1; fully functional cross-camera system delivered in FYP-2.)
- **Video Synopsis:** Condensing hours of surveillance footage into compact video summaries by extracting and temporally compositing only segments containing moving objects. (Processing begins in FYP-1; fully functional synopsis generation delivered in FYP-2.)
- **Vector Search & Storage:** Leveraging PostgreSQL with the pgvector extension to store embeddings and perform cosine similarity searches with IVFFlat/HNSW indexing for sub-100ms retrieval latency.
- **Shelf Engagement & Top-Selling Detection:** Analyzing customer dwell-time near product shelves using zone-based detection to identify high-engagement areas and top-selling product zones.
- **Fire & Smoke Detection:** Deploying a custom-trained YOLO model on fire/smoke datasets to provide real-time safety monitoring with instant alert notifications.
- **Crowd Density Analysis:** Estimating zone-level crowd density using person count and area-based thresholds to classify regions as crowded or uncrowded.
- **Checkout Counter Analytics:** Monitoring checkout lanes to track queue lengths, average service time per customer, and order throughput metrics.
- **Emotion Recognition:** Leveraging DeepFace/FER to detect customer sentiment (happy, neutral, frustrated, etc.) across store zones.
- **Secure Backend:** A FastAPI-based asynchronous backend that orchestrates the AI pipeline, manages JWT-based authentication, and handles SHA-256 hashing and AES-256 encryption.
- **Visualization Dashboard:** A React.js frontend with Recharts and Tailwind CSS presenting real-time dashboards for all analytics modules and security audit logs.

2. Success Criterion

The project will be deemed successful based on achieving the following measurable criteria:

Metric	Target	Method of Evaluation
Search Latency	< 100ms retrieval time	pgvector cosine similarity queries benchmarked on IVFFlat/HNSW indexes
Detection Accuracy	> 85% mAP	Evaluated on standard benchmarks (COCO, custom retail dataset) using YOLOv8/v11
Re-ID Accuracy	> 70% Rank-1	
Video Synopsis	>= 10x compression ratio	Hours of footage condensed to reviewable summaries preserving all moving-object events
Fire/Smoke Detection	> 90% precision, < 5% FPR	Custom-trained YOLO model evaluated on fire/smoke benchmark datasets
Crowd Classification	> 85% accuracy	Zone-level crowded/uncrowded classification tested against ground-truth annotations
Checkout Metrics	Queue length +/-1 person	Validated against manual counting on sample checkout footage
Emotion Recognition	> 70% accuracy	Tested on FER-2013 benchmark dataset using DeepFace/FER classifiers
Integrity Verification	100% tamper detection	SHA-256 chain verification detects any unauthorized database modifications
Throughput	20+ FPS	Multi-camera streams processed concurrently via FastAPI's async handling
Adversarial Robustness	Documented resilience	YOLO model evaluated against FGSM, PGD, and adversarial patch attacks using ART

3. Related Work

Person Detection & Tracking

Redmon et al. introduced the YOLO architecture, which revolutionized real-time object detection by framing it as a single regression problem [1]. The Ultralytics YOLOv8 further refines this with anchor-free detection heads [2]. Zhang et al. proposed ByteTrack for state-of-the-art multi-object tracking [3].

Person Re-Identification

Zhou et al. developed the Torchreid library for deep person re-identification with multiple backbone architectures [4]. Luo et al. proposed the 'Bag of Tricks' methodology for Re-ID [5]. The Market-1501 dataset by Zheng et al. established a standard benchmark [6].

Video Synopsis

Prabhakaran et al. introduced video synopsis as a method for condensing long surveillance videos into short, reviewable summaries by shifting detected activity tubes along the temporal axis [20]. Rav-Acha et al. formalized the synopsis optimization problem [21].

Vector Databases & Similarity Search

Johnson et al. at Facebook AI Research introduced FAISS for efficient similarity search [7]. The pgvector extension for PostgreSQL enables native vector operations with IVFFlat and HNSW indexing [8].

Adversarial Machine Learning

Goodfellow et al. introduced FGSM [9]. Madry et al. proposed PGD as a stronger adversarial attack [10]. The ART by Nicolae et al. provides a comprehensive framework for evaluating ML models against adversarial attacks [11].

Fire & Smoke Detection

Muhammad et al. proposed CNN architectures for early fire detection in surveillance videos [12]. Saeed et al. demonstrated that fine-tuning YOLO-based detectors on custom fire/smoke datasets yields robust real-time safety monitoring [13].

Crowd Density Estimation

Zhang et al. introduced CSRNet for crowd counting [14]. Sindagi and Patel surveyed CNN-based crowd counting methods for surveillance applications [15].

Emotion Recognition

Mollahosseini et al. developed AffectNet for facial expression recognition [16]. Serengil and Ozpinar introduced DeepFace, a lightweight framework for facial attribute analysis [17].

Retail Analytics & Computer Vision

Prior work has focused on customer behavior analysis using off-the-shelf trackers [18]. Recent studies explored shelf monitoring for planogram compliance [19]. OmniTrack advances this by integrating global Re-ID, video synopsis, multi-modal analytics, and a cryptographically secure audit trail.

4. Project Rationale

The motivation for OmniTrack AI stems from two converging industry needs:

- **Retail Intelligence Gap:** Physical retail stores generate enormous volumes of video data daily, yet this data is rarely leveraged for operational optimization. While e-commerce platforms track every click and scroll, brick-and-mortar stores lack equivalent digital insights. OmniTrack addresses this asymmetry by converting passive surveillance into active business intelligence.
- **Data Integrity & Security:** As AI-driven analytics systems become more prevalent in physical spaces, the integrity and security of the data they produce become paramount. Tampered analytics could lead to flawed business decisions. OmniTrack's Decoupled Security Pipeline ensures that every data point is cryptographically verifiable.

4.1 Aims and Objectives

Aim: To develop a secure, AI-driven multi-camera retail analytics platform that provides comprehensive operational intelligence -- including customer tracking, video synopsis, shelf analytics, safety monitoring, crowd management, checkout optimization, and emotion recognition -- while maintaining cryptographic data integrity.

1. Implement real-time person detection and multi-object tracking using YOLOv8/v11 and ByteTrack.
2. Deploy a global Re-Identification system using Torchreid to match individuals across multiple camera feeds.
3. Implement a Video Synopsis engine that condenses long surveillance recordings into compact, reviewable summaries.
4. Build a vector search infrastructure using pgvector for sub-100ms cross-camera identity matching.
5. Develop shelf engagement analytics to identify top-selling zones and high-interaction product areas.
6. Implement a fire and smoke detection module using custom-trained YOLO models.
7. Build a crowd density estimation module that classifies store zones as crowded or uncrowded.
8. Develop checkout counter analytics to track queue lengths, average service times, and order throughput.
9. Integrate a facial emotion recognition module using DeepFace/FER.
10. Design and implement a Decoupled Security Pipeline with SHA-256 hashing and AES-256 encryption.
11. Evaluate the detection model's resilience against adversarial ML attacks using ART.
12. Develop a React.js dashboard for real-time visualization of all analytics modules and security audit logs.

4.2 Scope of the Project

In Scope:

- Real-time person detection and tracking from multiple camera feeds.
- Cross-camera person Re-Identification using deep learning embeddings.
- Video Synopsis generation for rapid footage review and forensic analysis.
- PostgreSQL + pgvector based vector search and storage.
- Shelf engagement and top-selling product zone detection.
- Fire and smoke detection with real-time alerting.
- Crowd density estimation and crowded/uncrowded zone classification.

- Checkout counter queue length monitoring and order throughput analytics.
- Customer emotion recognition using DeepFace/FER.
- SHA-256 tamper-evident audit trail and AES-256 metadata encryption.
- Adversarial robustness evaluation of the YOLO detection model.
- FastAPI asynchronous backend with JWT authentication.
- React.js analytics dashboard with all modules and audit log viewer.
- Dockerized deployment with Nginx reverse proxy.

Out of Scope:

- Live deployment in an actual retail store (development uses benchmark datasets and simulated feeds).
- Point-of-Sale (POS) deep integration (checkout analytics are vision-based, not POS-integrated).
- Mobile application development.

4.3 FYP-1 / FYP-2 Development Phases

The project follows a two-semester development model:

FYP-1 (Current Semester - Spring 2026):

- Research, literature review, and environment setup.
- Core AI module development: person detection (YOLO), tracking (ByteTrack), Re-ID model training (Torchreid), and Video Synopsis processing pipeline.
- Extended analytics module development: shelf engagement, fire/smoke detection, crowd density, checkout analytics, emotion recognition.
- Initial security pipeline implementation (SHA-256, AES-256).
- Video Synopsis and Global Re-ID processing pipelines are built and validated individually but are not yet integrated into a fully functioning end-to-end system.
- Basic proof-of-concept dashboard demonstrating individual module outputs.

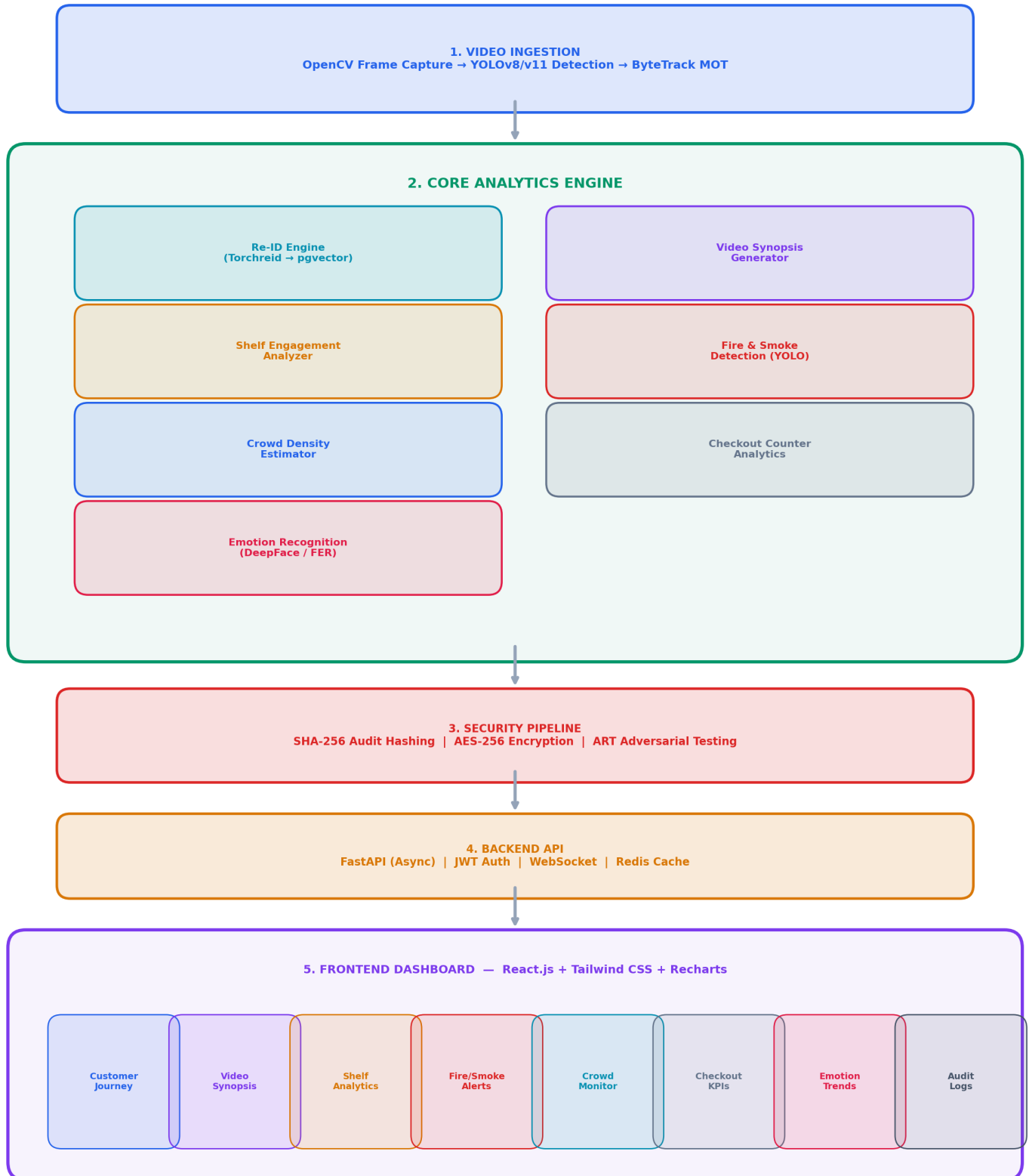
FYP-2 (Next Semester - Fall 2026):

- Full system integration: all modules connected via FastAPI backend with WebSocket real-time updates.
- Video Synopsis delivered as a fully functional feature with UI-based browsing, playback, and export.
- Global Re-ID delivered as a fully functional cross-camera tracking system with real-time identity matching and journey visualization.
- Complete React.js analytics dashboard with all modules, polished UI/UX, and production-ready performance.
- Comprehensive security pipeline with adversarial robustness testing (ART).
- End-to-end integration testing, performance benchmarking, and final documentation.

5. Proposed Methodology and Architecture

The system follows a layered pipeline architecture with the following distinct stages:

OmniTrack AI — System Architecture



5.2 Step-by-Step Procedure

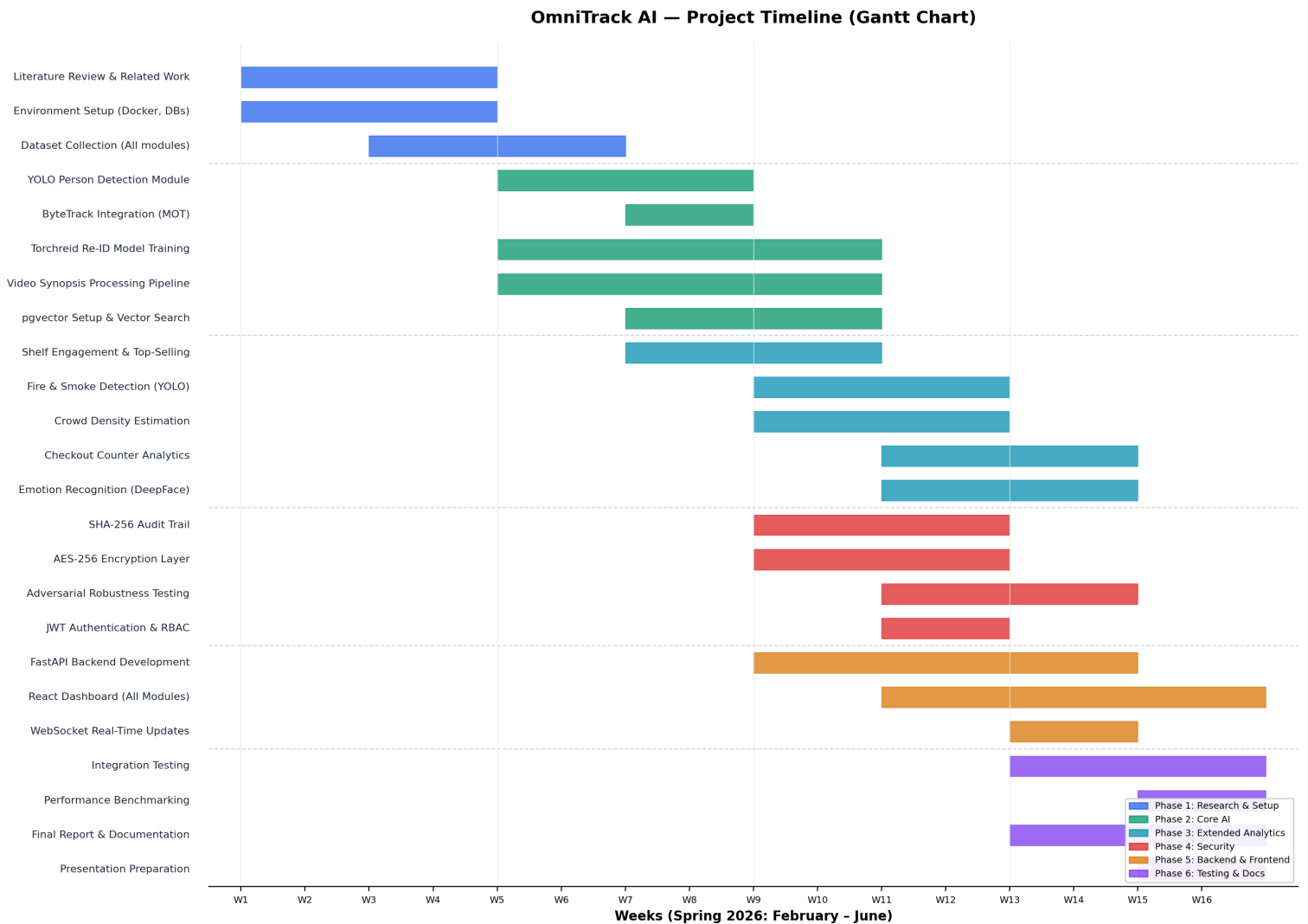
- **1. Ingestion:** Video frames are captured from multiple camera feeds via OpenCV and passed to the YOLOv8/v11 detector for real-time person detection.
- **2. Tracking:** ByteTrack assigns unique local IDs to detected persons within each camera's field of view, handling occlusions and re-entries.
- **3. Embedding Extraction & Re-ID:** The Torchreid model processes cropped bounding boxes and generates a 128-d or 512-d embedding vector. Embeddings are pushed to pgvector for cosine similarity search. (Processing built in FYP-1; fully integrated in FYP-2.)
- **4. Video Synopsis Generation:** Background subtraction isolates moving objects from static scenes. Activity tubes are extracted and temporally shifted to create a condensed video. (Processing pipeline built in FYP-1; full UI in FYP-2.)
- **5. Shelf Engagement Analysis:** Zone-based detection monitors customer positions relative to predefined shelf regions. Dwell-time is computed per zone to identify top-selling product areas.
- **6. Fire & Smoke Detection:** A parallel YOLO model, custom-trained on fire/smoke datasets, continuously scans camera feeds. Detections trigger instant WebSocket alerts.
- **7. Crowd Density Estimation:** Person count per defined zone area is computed in real-time. Zones are classified as crowded or uncrowded based on configurable density thresholds.
- **8. Checkout Counter Analytics:** Dedicated checkout cameras track queue length, average service time, and order throughput (customers served per hour).
- **9. Emotion Recognition:** Cropped face regions are passed to DeepFace/FER classifiers to identify emotions. Aggregated sentiment scores per zone feed into the analytics dashboard.
- **10. Security Layer:** FastAPI handles the request, hashes the transaction using SHA-256 to maintain a tamper-evident chain, and stores encrypted metadata using AES-256.
- **11. Visualization:** The React.js frontend displays real-time dashboards covering all analytics modules, video synopses, and a security audit log viewer.

6. Individual Tasks

Team Member	Activity	Tentative Date
Zeerak Shahzad (22K-4692)	Literature Review & Environment Setup	Week 1 - Week 4
	YOLO Detection & ByteTrack Integration	Week 5 - Week 8
	FastAPI Backend Development	Week 9 - Week 14
	Integration Testing & Final Report	Week 13 - Week 16
Member 2 (Reg #)	Dataset Collection & Preprocessing	Week 3 - Week 4
	Torchreid Re-ID + Video Synopsis	Week 5 - Week 10
	React Frontend & Dashboard Development	Week 11 - Week 14
	Performance Benchmarking	Week 15 - Week 16
Member 3 (Reg #)	pgvector Setup & Vector Search Tuning	Week 7 - Week 10
	SHA-256 / AES-256 Security Pipeline	Week 9 - Week 12
	Adversarial Robustness Testing (ART)	Week 11 - Week 14
	Presentation Preparation	Week 15 - Week 16

7. Gantt Chart

Session: SPRING 2026 (16 Weeks: February - June 2026)



8. Tools and Technologies

Layer	Technologies	Purpose
Vision Engine	Ultralytics (YOLOv8/v11), PyTorch, Torchreid, OpenCV	Person detection, tracking, Re-ID, fire/smoke detection, video synopsis
Emotion Analysis	DeepFace, FER	Customer sentiment classification from facial expressions
Vector Database	PostgreSQL with pgvector	Embedding storage and cosine similarity search (IVFFlat/HNSW)
Backend API	FastAPI (Async Python)	RESTful endpoints, WebSocket, AI pipeline orchestration
Frontend	React.js, Tailwind CSS, Recharts	Multi-module analytics dashboard
Security	PyCryptodome, JWT, ART	Data encryption, audit trail integrity, adversarial model evaluation
DevOps	Docker, Redis, Nginx	Containerization, caching, reverse proxy
Development	Git, GitHub, VS Code, Python 3.11+	Version control, IDE, scripting

9. References

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Supervisor Approval

	Name	Signature	Date
Supervisor			
Co-Supervisor			